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System boundary

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System: Inspection Report Routing System

Inside the boundary

The system receives inspection reports from the field, checks required fields against contracts, looks up routing destinations, creates and sends work items to receiving departments, and logs report records and delivery statuses.

Outside the boundary

Field operations such as drone flying and physical inspections, physical maintenance and repair work, flight control systems, and service administration or table maintenance done outside this system.

Boundary flows

One row per information flow. Repeat an entity if it has more than one flow.

External entity	Direction (into / out of system)	Data flow name	Source
Field Team	into system	Inspection report	W06-S01
Field Team	out of system	Validation note	W06-S01
Electrical	out of system	Routed work item	W06-S01.W06-S03
Drainage	out of system	Routed work item	W06-S01.W06-S03
Structures	out of system	Routed work item	W06-S01.W06-S03

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Information movement

Processes: 1 Receive Report; 2 Check Required Fields; 3 Resolve Destination; 4 Send Work Item. Stores: D1 Report Register; D2 Routing Table. Use the flow rows to connect these and the external teams from page 1. Eleven flow rows are provided.

From (entity / process / store)	To (entity / process / store)	Data flow name	Source
Field Team	Receive Report	Inspection report	W06-S01
Receive Report	D1 Report Register	Submitted report	W06-S01
Receive Report	Check Required Fields	Received report	W06-S01
Check Required Fields	Field Team	Validation note	W06-S01
Check Required Fields	Resolve Destination	Accepted report	W06-S01
Resolve Destination	Resolve Destination	Routing entries	W06-S01
D2 Routing Table	Send Work Item	Report with resolved destination	W06-S01
Send Work Item	Electrical	Work item	W06-S01, W06-S03
Send Work Item	Drainage	Work item	W06-S01, W06-S03
Send Work Item	Structures	Work item	W06-S01, W06-S03
Send Work Item	Report Register	Delivery record	W06-S01, W06-S04

Balance check:

External inputs consisting of inspection reports from the Field Team and external outputs consisting of validation notes back to the Field Team and work items sent to Electrical, Drainage, and Structures match the boundary data flows on Page 1. Every internal process receives and outputs data correctly.

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Data definitions and finding

Field	Meaning	Example	Allowed value / format	Source
report_id	Unique report reference	CJ-601	CJ- followed by three digits	W06-S02
asset_id	Reference for the inspected asset	SL-101	A two-letter prefix and three digits	W06-S02
asset_type	Functional classification of the inspected asset	STREETLIGHT	One of STREETLIGHT, DRAIN or BRIDGE	W06-S02
finding_code	Standardized code for condition or malfunction detected	LAMP_OUT	One of LAMP_OUT, BLOCKED or SURFACE_DAMAGE	W06-S02
priority	Urgency level for follow-up resolution	ROUTINE	One of ROUTINE or URGENT	W06-S02
captured_at	UTC capture time of the inspection record	2026-10-14T09:00:00Z	UTC capture time written as YYYY-MM-DDTHH:MM:SSZ	W06-S02
photo_ref	Fictional attachment reference for inspection images	IMG-601	IMG- followed by three digits	W06-S02
survey_area	Fictional geographic or operational survey zone	NORTH	One of NORTH, CENTRAL or SOUTH	W06-S02

Supported finding

Report `CJ-603` has an `asset_type` of `BRIDGE`, which policy (W06-S03) states belongs to Structures. However, the delivery log (W06-S04) records that `CJ-603` was actually received by Electrical (W06-S02, W06-S03, W06-S04).

Unanswered question:

What specific routing table configuration or table entry caused bridge reports to be sent to Electrical instead of Structures (W06-S03, W06-S04)?